CHARACTERIZATION OF RIGHT ACYCLIC OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5849-5858, following the statement at 5835-5847.

Editorial note. No omissions: both directions are executed; the converse is a direct application of the named lemma.

Allowed standard prerequisites (editorial).

- Cohomology of the derived functor. lemma-left-exact-higher-derived (carded L26184, cited at ~5855). Part (1) gives the quasi-isomorphism; also F = R^0F for left exact F.

Source reference. Original lemma 05TD: sources/stacks-derived/pinned-original-source.tex, source lines 5782–5782. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor between abelian categories and assume $RF : D^+(\mathcal{A}) \to D^+(\mathcal{B})$ is everywhere defined. Let A be an object of $\mathcal{A}$.*

(1) *A is right acyclic for F if and only if $F(A) \to R^0F(A)$ is an isomorphism and $R^iF(A) = 0$ for all $i > 0$,*
(2) *if F is left exact, then A is right acyclic for F if and only if $R^iF(A) = 0$ for all $i > 0$.*

Proof. If A is right acyclic for F, then $RF(A[0]) = F(A)[0]$ and in particular $F(A) \to R^0F(A)$ is an isomorphism and $R^iF(A) = 0$ for $i \neq 0$. Conversely, if $F(A) \to R^0F(A)$ is an isomorphism and $R^iF(A) = 0$ for all $i > 0$ then $F(A[0]) \to RF(A[0])$ is a quasi-isomorphism by Lemma 05TD part (1) and hence A is acyclic. If F is left exact then $F = R^0F$, see Lemma 05TD. □

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.